

Thrust Chamber Assembly Design – YJSP

BOOTCAMP FALL 2025

BACKGROUND

Liquid rocket engines (LREs) utilize chemical combustion, a process which releases abundant thermal and kinetic energy accelerating the breaking of molecular bonds, to generate thrust for a wide variety of applications, from launch vehicles to in-space maneuvering. In fact, burning a single kilogram of hydrogen gas in oxygen gas is roughly the same as setting off 35 kgs of TNT. Now imagine hundreds of kilograms of fuel and oxidizer combusting every second in the same volume as a car!

Of course, without proper control over the combustion reaction, the thrust that is generated will be very little, creating a rocket engine that is very inefficient. This is exactly the purpose of an engine thrust chamber assembly (TCA): to control and the direct hot combustion gases, allowing thermal energy to be converted into bulk momentum.



Thrust Chamber Assemblies from Rocket Lab: RP-1/LOX Rutherford (left) and MethaLOX Archimedes (right)

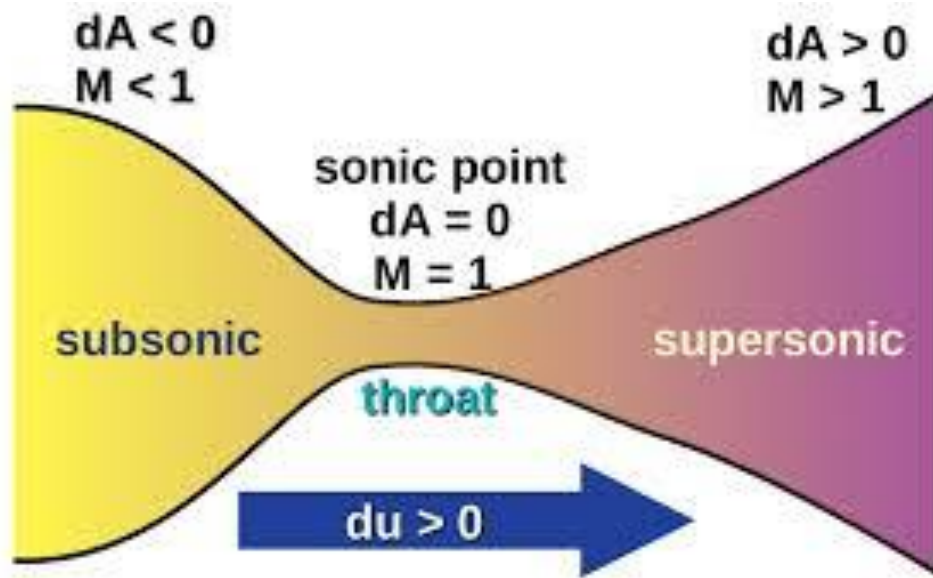
When designing a TCA for a bipropellant LRE, the internal shape, or contour, is highly dependent on the propellants that are being used. The pressure in the combustion chamber and the relative ratio of the propellant masses (mixture ratio) directly determine how much energy is available for conversion into kinetic. With these parameters, the combustion chamber and nozzle must be designed to provide the *optimal* balance between efficient thrust production, weight, and envelope (how much space the engine takes up). Therefore, the geometry of a TCA is critical to the efficiency of the overall vehicle.



Different TCAs used for different applications: Raptor (left) vs Raptor Vacuum (right)

Most rocket nozzles today rely on **compressible flow theory** to predict the thrust generated by a propellant combination. That is, because the substances inside a TCA are often high-temperature and gaseous, we cannot rely on the assumption that the density of the fluid will remain constant as the flow propagates from the chamber to the exit. Additionally, as a form of baseline analysis, engineers typically assume that the flow inside of a rocket nozzle is *steady* and *isentropic* (change in entropy is zero). That is, the mass flow entering the nozzle is the same mass flow that exits the nozzle and the flow itself is 100% efficient: all the energy released from splitting molecules is converted into kinetic energy (if you are interested in learning more about entropy, please come ask me! I think it's a really cool concept that most people don't understand very well). Of course, these assumptions aren't always true. For example, upon engine startup, the mass of propellants entering the chamber is not matched by the exit flow. The flow is also never 100% efficient; heat transfer, shocks, and chemical kinetics are just some of the sources

of inefficiency in these systems, but analysis of these phenomena is non-trivial and, oftentimes, very different from reality.



Choked flow in a De Laval nozzle: one of the many applications of compressible flow theory

In this project, you will be designing your own TCA contour for liquid bipropellant engine that uses RP-1 and LOX (liquid oxygen), which are the propellants that YJSP's KeroLOX vehicles use. Given a set of design requirements, you are asked to calculate certain fluid properties of the combustion gases in the TCA, conceptualize the internal shape of the TCA, and establish the effect of the real losses on the overall efficiency of your TCA. In doing so, you will understand the major design, performance, and geometric parameters that engineers use to qualify engine combustion processes and TCA contours. Additionally, you will develop an understanding of the design process behind Vesputa's engines.

Typically, this process is highly iterative as the design of the TCA is intertwined with the designs of the feed system, pumps, and injectors. Moreover, TCAs are often limited by size, weight, and manufacturing constraints. Here, you will do a base level analysis ignoring those considerations for now. If you are interested, you are free to discuss how these constraints would affect your engine's parameters, but the primary focus will be on combustion analysis and contour design.

This project is designed to be done in conjunction with the Regen TCA design project (TCA Design Part II), so the contour you design can be directly implemented in that project so that you can learn from the full design process. This is meant to be open ended so that you can collaborate with other people and see that not every problem has one solution.

OBJECTIVE/REQUIREMENTS

- Design a TCA contour for an RP-1/LOX engine and estimate the performance of the TCA for a nozzle designed with the following parameters
 - Chamber Pressure: 300 psia
 - Exit-plane Pressure: 10 psia
 - Thrust: 2500 lbf
 - Assume C^* Efficiency of 92%
- For your equilibrium model, use the following fluid properties:
 - The reaction is adiabatic (when accounting for enthalpies of formation)
 - LOX Propellant Temperature: 90.170 K
 - RP-1 Propellant Temperature: 298.15 K
 - RP-1 (Rocket Propellant 1) is actually a kerosene mixture with hundreds of hydrocarbon types. Therefore, it is hard to model as a single fluid. You may use n-Dodecane (or just Dodecane) as a surrogate fluid, as it has readily available data in CoolProp or NIST
 - You can also use Jet-A (l) if you're using Thermobuild
 - Enthalpies of formation (all in J/kg):
 - O_2 : 0.0
 - $C_{12}H_{26}$: -1.45×10^5
 - CO : -3.95×10^6
 - CO_2 : -8.94×10^6
 - H_2O : -1.34×10^7
 - H_2 : 0.0
 - OH : 2.29×10^6

- This project involves two distinct parts: making an equilibrium script and designing a TCA contour. If you're interested in doing only one of those parts, that's completely fine! Either way, I will be open to helping you. Again, keep in mind that this is meant to be relatively open-ended project where you learn as much as you want to.

DELIVERABLES

- Design a very basic chemical equilibrium calculator for the RP-1/LOX reaction:
 - o You can go into as much detail as you want on this, but I recommend sticking to the Major-Minor species model (so no K_p equations, only linear equations with atom balances):
 - Equivalence ratio for chosen mixture ratio
 - Table comparing mole (or mass) fractions of species concentrations between your model and CEA (assuming frozen flow)
 - Discussion on why the concentrations vary between your model and CEA. How can you make your model more accurate?
 - Table comparing the following quantities between your model and CEA (assuming frozen flow):
 - Adiabatic Flame Temperature (K)
 - Chamber Molecular Weight
 - Chamber Specific Heat Ratio
 - Exhaust Mach Number
 - Exhaust Temperature (K)
 - Exhaust Velocity (m/s)
 - Specific Impulse (s)
 - Discussion on why the frozen flow assumption needs to be made for CEA to compare to your model
 - Discussion on how much your model varies from CEA and why. What are some improvement techniques? What are the shortcomings of your model? How could you extend it to become a general equilibrium solver?
- Your design should be comprehensive and optimized, and should include the following outputs:

- You should use an actual chemical equilibrium tool for this part
- Design variables:
 - Mixture Ratio
 - NFZ
 - Discussion on why you chose that mixture ratio and how. Why did you choose that NFZ model?
- Performance variables:
 - Specific Impulse (s) (vacuum and ambient)
 - Characteristic Velocity (m/s)
 - Thrust Coefficient
 - Discussion on what each of these variables represents and why they're important. How can each variable be optimized?
- Geometric Variables:
 - Summarize relevant geometric properties in a table
 - Use them to create a figure of your contour
 - You will be provided with a contour generation script
 - **(Bonus)** Make your own contour generation script
 - Discussion on how and why you chose each geometric parameter.
- Efficiencies **(Bonus)**:
 - These will all be estimates since it is hard to exactly determine efficiencies without testing
 - Rayleigh Line Loss
 - Discharge Coefficient
 - Energy Release Efficiency
 - Nozzle Efficiency
 - Discussion on how you would optimize each of these variables

CHECKPOINTS

1. Kickoff
 - a. Discuss the purpose of nozzle optimization, chemical equilibrium analysis, and performance calculations
2. Chemical Equilibrium Analysis
 - a. Check to make sure the code is working as expected
 - b. Troubleshoot any conceptual or script issues
3. MR Selection
 - a. Once your mixture ratio is selected, we can make sure your methodology is reasonable.

4. Contour Design
 - a. We will check in to make sure your contour makes sense and is suitable for the given application
5. Efficiencies (Bonus)
 - a. We will discuss how to find efficiencies and how to implement the code

RESOURCES

- Saaketh Ramoju
- *Rocket Propulsion Elements* – George P. Sutton
- *How to design, build and test small liquid-fuel rocket engines* - <https://spacha.github.io/How-to-Rocket/#design-equations>
- *Design of Liquid Propellant Rocket Engines* – Dieter K. Huzel and David H. Huang
- NASA CEA - <https://cearun.grc.nasa.gov/>
 - This is the industry standard chemical equilibrium tool for TCAs
 - NASA CEA provides quick and simple chemical analysis online
- RocketCEA - <https://rocketcea.readthedocs.io/en/latest/index.html>
 - A python wrapper around NASA CEA
 - You can install it by installing python (version 3.0 and above) and running the following in your terminal/command prompt:
 - `pip3 install numpy matplotlib scipy rocketcea`
 - This will include rocketcea as well as all supporting packages
 - Read the documentation here: <https://rocketcea.readthedocs.io/en/latest/functions.html>
- CoolProp - <https://coolprop.org/index.html>
 - A general coding package for fluid properties (use the Python version)
 - Very commonly used in industry for real fluid properties
 - USE THIS FOR YOUR EQUILIBRIUM SCRIPT
 - Run the following in terminal/command prompt to install:
 - `pip3 install CoolProp`
 - If you're using the PyFluids wrapper:
 - `pip3 install pyfluids`
- RocketIsP - <https://rocketisp.readthedocs.io/en/latest/index.html>
 - This tool allows for basic efficiency calculations and geometry generation
 - Can also gather performance metrics from this
- NASA Thermobuild - <https://cearun.grc.nasa.gov/ThermoBuild/>

- The fluid properties backed for NASA CEA
 - Use this to get gas properties for CEA calculations
- NIST Thermophysical Webbook - <https://webbook.nist.gov/chemistry/fluid/>
 - Use this to quickly lookup pure fluid properties
 - Be careful this for mixture properties
- NIST – JANAF Tables - <https://janaf.nist.gov/>
 - This is a better version of NASA Thermobuild
 - Provides values that are necessary for chemical equilibrium calculations like enthalpies of formation and equilibrium constants
- Interactive Engine Design Guide – <https://github.com/saakethramoju/Engine-Design-Guide>
 - This will be your best resource by far
 - There are many resources listed at the bottom of this guide by the way
 - This also includes a contour design applet at the bottom
- Contour Design Widget – <https://saakethramoju.github.io/Contour-Generator/>
- *ROCKET PROPULSION* - <http://www.braeunig.us/space/propuls.htm>
- *CRYO-ROCKET* - <https://www.cryo-rocket.com/>
- AE 2010 Professor Seitzman's Slides: <https://seitzman.gatech.edu/classes/ae2010/>
- AE 6450 Professor Seitzman's Slides: <https://seitzman.gatech.edu/classes/ae6450/>
- AE 6765 Professor Seitzman's Slides: <https://seitzman.gatech.edu/classes/ae6765/>
- AE 4451 Professor Seitzman's Slides: <https://seitzman.gatech.edu/classes/ae4451/>